The Power of Symmetry

Over the years, no one even attempted to study the properties of any of the five string theories for large values of their string coupling constants because no one had any idea how to proceed without the perturbative framework. However, during the late 1980s and early 1990s, physicists made slow but steady progress in identifying certain special properties—including certain masses and force charges—that are part of the strong-coupling physics of a given string theory, and yet are still within our ability to calculate. The calculation of these properties, which necessarily transcends the perturbative framework, has played a central role in driving the progress of the second superstring revolution and is firmly rooted in the power of symmetry.

Symmetry principles provide insightful tools for understanding a great many things about the physical world. We have discussed, for instance, that the well-supported belief that the laws of physics do not treat any place in the universe or moment in time as special allows us to argue that the laws governing the here and now are the same ones at work everywhere and everywhen. This is a grandiose example, but symmetry principles can be equally important in less all-encompassing circumstances, For instance, if you witness a crime but were able to catch only a glimpse of the right side of the perpetrator’s face, a police artist can nonetheless use your information to sketch the whole face. Symmetry is why. Although there are differences between the left and right sides of a person’s face, most are symmetric enough that an image of one side can be flipped over to get a good approximation of the other.

In each of these widely different applications, the power of symmetry is its ability to nail down properties in an indirect manner—something that is often far easier than a more direct approach. We could learn about fundamental physics in the Andromeda galaxy by going there, finding a planet around some star, building accelerators, and performing the kinds of experiments carried out on earth. But the indirect approach of invoking symmetry under changes of locale is far easier. We could also learn about features on the left side of the perpetrator’s face by tracking him down and examining it. But it is often far easier to invoke the left-right symmetry of faces.7